Iconicity in word order variation

The case of preposition placement

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Introduction

(1) I rely **on** findings **from** the literature (BNC)

(Huddleston & Pullum, 2002, pp. 627–631)

Introduction

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(2) a. the sources **on** which they rely (BNC)

b. the sources which they rely **on**

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b. the sources which they rely **on**

(3) a. *[he] slept badly, for no reason **of** (that) he knew (BNC)

b. [he] slept badly, for no reason (that) he knew **of**

(Huddleston & Pullum, 2002, pp. 627–631)

Introduction

Table: Distribution of prepPlace by clause type in English (ICE-GB)

	Clause type					
PrepPlace	wh-RC	Non-wh-RC	Free RC	PASS	Wh-Q	Total
Fronting	692	0	4	0	12	708
Stranding	69	419	157	97	140	882
Total	761	419	161	97	152	1590

Note. RC = relative clause, PASS = passive clause, Q = question clause.
Adapted from Hoffmann (2011, pp. 120 & 158).

Introduction

Specific lexical items and strings
Salience and input frequency
Meaning of the filler item
Cross-linguistic similarity
Information packaging
Individual speakers
Clause complexity
Clause meaning
Clause type
Modality
Register
Style

Bardovi-Harlig (1987), Bergh and Seppänen (2000), Biber, Johansson, Leech, Conrad, and Finegan (1999), Van Buren and Sharwood Smith (1985), Chomsky (1981, 1995), Dekeyser (1990), Gries (2002), Guy and Bayley (1995), Hamilton (1995), Hawkins (1999), Hoffmann (2005, 2006, 2007, 2008, 2011, 2013), Hornstein and Weinberg (1981), Huddleston and Pullum (2002), Hudson (2018), Johansson and Geisler (1998), Kao (2001), McDaniel, McKee, and Bernstein (1998), Van Riemsdijk (1978), Ross (1967), Snyder (2007), Sugisaki and Snyder (2006), Stowell (1981), Takami (1992), Trotta (2000), Yáñez-Bouza (2015), among others

Introduction

Question What determines the variation between preposition fronting and stranding in relative clauses in English as a second language?

Method A corpus study was conducted to analyze prepPlace in English language use by nonnative learners compared to native speakers.

Results Nonnative prepPlace depends (a) on proficiency, (b) on specific lexical items and strings, (c) on cross-linguistic similarity, and **(d) on the meaning of the filler item**.

Discussion This suggests that nonnative prepPlace is acquired in a usage-based manner and affected by crossmodal processes like analogy (e.g., Diessel, 2017; Ellis, Römer, & O'Donnell, 2016). In particular, adopting a cognitive grammar framework (Langacker, 2008a), I argue that the meaning effect arises from iconicity.

Overview

1. Introduction

2. The Current Study
2.1 Method
2.2 Results

3. Discussion

4. Limitations

Section 2

The Current Study

To examine prepPlace in nonnative English language use, a sample of relative clauses from English native and nonnative corpora was subjected to a multivariate logistic regression analysis predicting prepPlace from multiple predictor variables.

Jach, D. (2018). *Something I was dealing with: Preposition Placement in Multilingual Constructicons*. [WT] Manuscript submitted for publication.

Corpora and Sample

	Learner group		Proficiency	
Corpora	Native	Nonnative	Novice	Advanced
ICE-GB (written)	×			×
LCN	×		×	
ICLE		×	×	
YELC (B1–C2)		×	×	

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	L1s							
PrepPlace	Eng	Ger	Fre	Ita	Spa	Chi	Jap	Kor
Fronting	×	×	×	×	×			
Stranding	×							

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	L1s							
PrepPlace	Eng	Ger	Fre	Ita	Spa	Chi	Jap	Kor
Fronting	×	×	×	×	×			
Stranding	×							

	PrepPlace		
Learner group	Fronting	Stranding	Total
Native	628 (628)	250 (41)	878 (669)
Nonnative	853 (852)	567 (116)	1420 (968)
Total	1481 (1480)	817 (157)	2298 (1637)

Note. Frequency of *wh*-RCs in parentheses.

Variables

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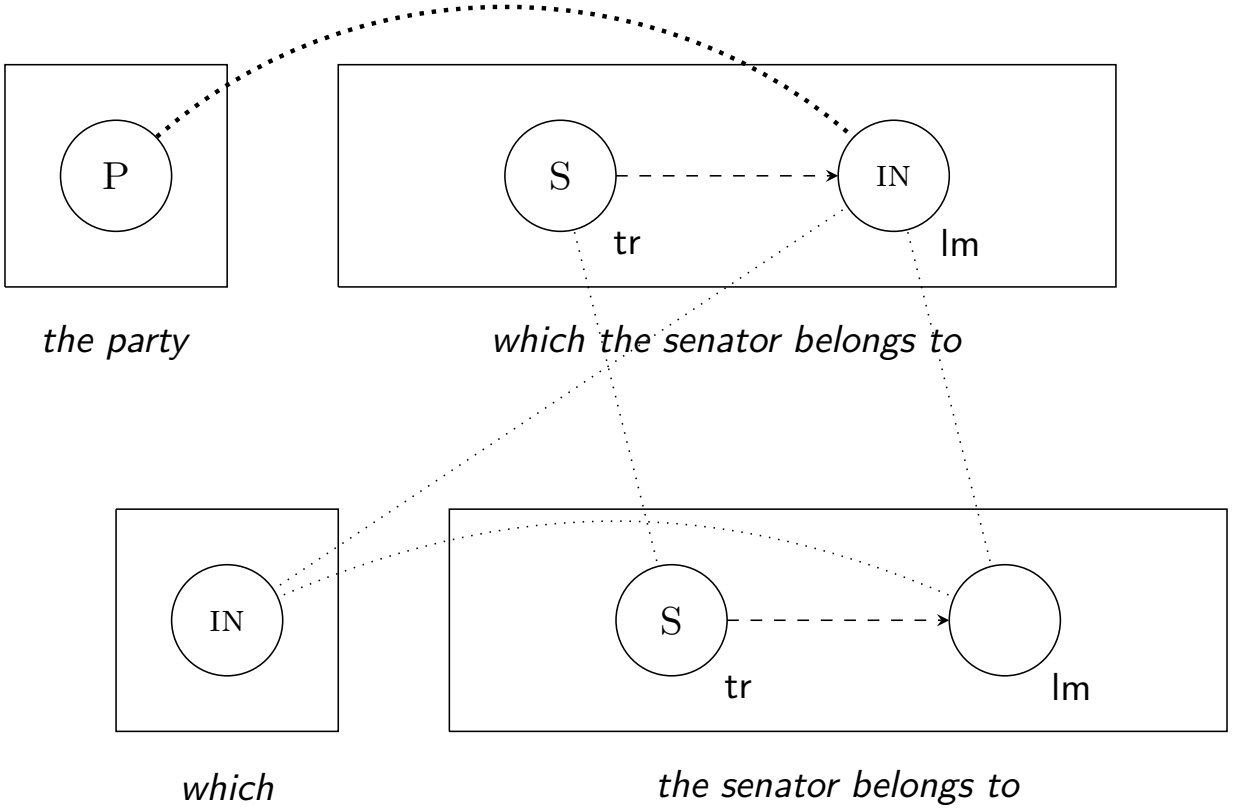

Variables

- Learner group + proficiency
- Meaning of the filler item

Linguistic expressions evoke organized packages of knowledge as their conceptual background of understanding, so-called *frames* (Fillmore, 1985).

(4) The soda which I **bought** for five dollars from the supermarket downtown yesterday tasted like water.

Variables

- Learner group + proficiency
- Meaning of the filler item

(4) Participant
the party *which the senator* ***belongs to*** (ICE)

(5) Supplement
[t]he ***method*** *in which to* ***carry out*** *this type of torture* (ICLE)

(6) Environment
the ***world*** *in which I have* ***grown up*** (ICE)

(Hoffmann, 2011; Hornstein & Weinberg, 1981; Ruppenhofer, Ellsworth, Petruck, & Johnson, 2016)

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L1 European (German, French, Italian, Spanish)

(7) *die Münze,* ***von*** *der ich (*****von****) gesprochen habe*
the coin of which I of spoken have

"the coin about which I talked" (Eisenberg et al., 2009, p. 1030)

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L1 East Asian (Chinese, Japanese, Korean)

(8) *wǒ (******yòng****) xiūlǐ shuǐ guǎnzi (******yòng****) de jùzi*
I with repair water pipe with REL saw

"the saw with which I repair the water pipe"
(Li & Thompson, 1981, p. 582)

Variables

- Learner group + proficiency
- Meaning of the filler item
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- Specific lexical items and strings and their usage frequency

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(9) a. the car he left his coat **in** (Quirk et al., 1985, p. 664)

b. Who are you **looking for**? (Biber et al., 1999, p. 106)

c. the way **in which** he did it (Hoffmann, 2011, p. 22)

Results

Table: Binary mixed-effects regression model predicting prepPlace from multiple variables

	b	*SE*	*z*	*p*
Intercept	-5.92	1.6	-3.71	***
European L1	-1.34	0.64	-2.1	*
East Asian L1	1.1	0.73	1.5	ns
Participant filler	2.27	0.77	2.95	**
Supplement filler	-0.83	0.99	-0.84	ns
Novices (LCN, ICLE, YELC)	1.83	0.77	2.38	*
that	9.1	2.39	3.8	***
what/whatever	5.72	1.7	3.36	***
where	1.67	1.47	1.14	ns
whom/who/whose	-2.04	0.96	-2.13	*
Frequency preposition	1.36	0.61	2.22	*
Usage frequency preposition-item string	-1.53	0.56	-2.75	**
Usage frequency item-preposition string	1.06	0.32	3.29	**

Note. ns $p > .10$, . $p \leq .10$, * $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$.

Results

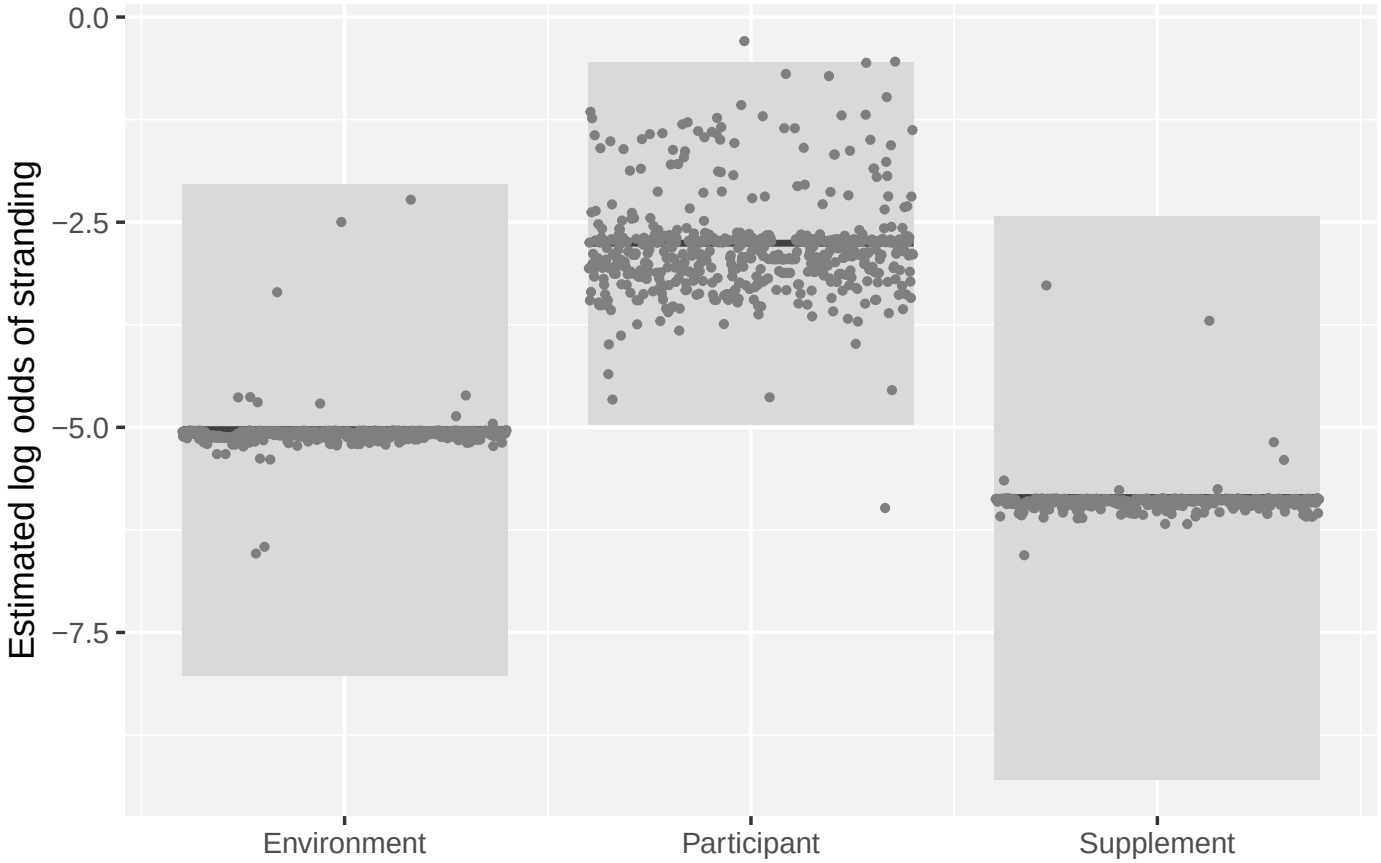

Table: Part of a binary regression model including L1 and Proficiency as predictors

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Summary of Results

Question What determines the variation between preposition fronting and stranding in relative clauses in English as a second language?

- Meaning of filler items
- Proficiency
- Cross-linguistic similarity
- Specific lexical items and strings

Section 3

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(10) a. the boat on which John decided

b. the boat which John decided on

(Adapted from Hornstein and Weinberg, 1981)

Discussion

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Behaghel's "first law"

> *Geistig eng Zusammengehöriges wird auch eng zusammengestellt.* [Conceptually related entities are placed close to each other.] (Behaghel 1932)

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Table 7.4. Order of premodifier adjectives

functions:		**grounding**	**characterising**	**categorising**	
(a)		determining	scalar	denominal	
	the	*only*	*reliable*	*economic*	*expert*
(b)		determining	deadverbial	denominal	
	the	*first*	*intelligent*	*diplomatic*	*solution*

(11) a. stupid white man

b. pretty young girl

(Radden & Dirven, 2007, p. 154)

Discussion

In prepPlace, since the filler nominal is immobile, the mobile preposition "takes over the job." Different word orders invoke different modes of mental scanning during integration: summary (a) vs. sequential (b), resulting in different readings.

(10, repeated) a. **the boat** on which John **decided**

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(Adapted from Hornstein and Weinberg, 1981)

Grammar is meaningful (e.g., Langacker, 2008a, 2015; Talmy, 1988).

Discussion

With nonverbal frames, however, prepositions tended to front, independent of filler meaning.

(12) a. **an economic challenge** of which **the aim** is to build... (ICLE)

b. ***an economic challenge** which **the aim** of is to build...

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Fronting construes the filler as out of focus, stranding as in focus *at the level of the RC*.

(13) a. **AIDS**, for which we have **no cure** (LCN)

b. **AIDS**, which we have **no cure** for

Section 4

Limitations

Limitations

Need for improvement

- While "the mere existence of sequential and summary scanning as conceptual phenomena is hardly problematic" (Langacker, 2008b, p. 572), empirical evidence for their role in language processing is sparse (but see Hoffmann, 2011; Matlock, 2004)
- More reliable, fine-grained measure of proficiency
- More control of register, style, and other confounding cross-corpus differences

Jach, D. (2018). A usage-based approach to preposition placement in English as a second language. *Language Learning, 68*, 271–304.
https://doi.org/10.1111/lang.12277

Thank you for your attention.

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